

Stopped Channels and Offset Alignment: De-Conditionalizing the Real-Channel Hypotheses for Mixed-Base Tetration

Paper III of the mixed-base tetration series

Janis Justus

July 2026

Abstract

The mixed-base phase law of [1] and its diffeomorphism companion [2] are conditional on two hypotheses about the regular branch: the real-channel positivity (R2)/(R') and the peeling-growth hypothesis 2.4. We remove most of this conditionality. (i) The growth hypothesis holds unconditionally for every admissible tower base $d > e^{1/e}$, with the explicit gap $T_d(j+1) - T_d(j) \geq (1 + \ln \ln d) / \ln d > 0$. (ii) Positivity of the finite logarithmic chain holds unconditionally down to an explicit *stop level* $J_0 = \min\{j : T_d(j) \geq 2C_0\}$ (the *Stoppregel*), by a self-contained downward induction that produces, rather than assumes, the channel; the evaluation and limit machinery of [1, 2] only ever needs the stopped chain. (iii) For the limit objects the correct channel statement is *offset-aligned*: $E_j = T_b(j + \tilde{H}(\theta)) > 0$ is required only for $j \geq s$, where $s = \lceil -1 - \inf \tilde{H} \rceil^+$ is computable a priori from the mean-shift hub; the unshifted reading is false for pairs with $\mu_{b,d} \leq -2 + \|P\|_\infty$ — e.g. $(b, d) = (10, 2)$ and $(e, 1.5)$ — while the offset version holds with explicit margins. A numerical certification across all 45 measured pairs is reported. Together with the base-differentiability of the Kneser family established by the response-kernel program, the conditional core of the series reduces to a single mode-certifiable inequality per pair.

1 Introduction

The phase law of Paper I [1] and its diffeomorphism companion, Paper II [2], are conditional on two hypotheses about the regular Kneser/Paulsen–Cowgill branch: positivity of the real logarithmic channel ((R2) of [1], (R') of [2]) and the peeling-growth hypothesis (Hyp. 2.4 of [1]). This paper removes most of that conditionality, in the form in which the estimates of the series actually consume it.

The contributions are: (i) the growth hypothesis holds *unconditionally* for every admissible tower base $d > e^{1/e}$, with the explicit gap $d^x - x \geq (1 + \ln \ln d) / \ln d > 0$ (Section 3); (ii) positivity of the finite logarithmic chain holds unconditionally down to an explicit stop level J_0 , by a self-contained downward induction whose invariant is at the same time the error control of the evaluation ladder used throughout the series (the *Stoppregel*, Section 4); (iii) for the limit objects, the correct channel statement is *offset-aligned*: positivity is required only from an explicit shift $s = \lceil -1 - \inf \tilde{H} \rceil^+$ onward, computable a priori from the mean-shift hub, and the unshifted reading is provably false for skew pairs (Section 5); (iv) the alignment margins are certified numerically across 45 measured pairs (Section 6). Together with the base-differentiability theorem of Paper V [3], the conditional core of the series reduces to a single mode-certifiable inequality per pair.

2 Setting

We keep the conventions of [1, 2]. For an admissible base $a > e^{1/e}$, T_a denotes the regular (Kneser/Paulsen–Cowgill) tetration branch with $T_a(0) = 1$, $T_a(x+1) = a^{T_a(x)}$, strictly increasing from $(-2, \infty)$ onto $\mathbb{R}$, and $A_a = T_a^{-1} = \text{slog}_a$. For an ordered pair (b, d) set

$$\alpha = \log_b d, \quad L_b = \log_b, \quad s_j = \inf_{0 \leq \theta \leq 1} T_d(j + \theta) = T_d(j),$$

$$C_0 = \frac{|L_b(\alpha)| + \ln 2 / \ln b}{\alpha}, \quad W_k^{(n)}(\theta) = L_b^{\circ k}(T_d(n + \theta)).$$

$\Phi_{b,d}$, $\mu_{b,d} = \int_0^1 \Phi_{b,d}$, $P_{b,d} = \Phi_{b,d} - \mu_{b,d}$ and $\tilde{H}_{b,d} = \text{id} + \Phi_{b,d}$ are as in [1, 2], and $A_k = A_k(b, d)$ denotes the Fourier amplitudes of $P_{b,d}$.

Hypothesis (R') of [2] (the quantitative reading of (R2) in [1]) demands $W_k^{(n)} > 0$ for $0 \leq k \leq n-1$. Hypothesis 2.4 of [1] demands $s_j \rightarrow \infty$. Both were left conditional in [1, 2] (see [2, Rem. 7.4]).

3 The growth hypothesis is free

Lemma 1 (Tower divergence with explicit gap). *Let $d > e^{1/e}$. Then for all $x \in \mathbb{R}$,*

$$d^x - x \geq \delta_0(d) := \frac{1 + \ln \ln d}{\ln d} > 0.$$

Consequently $s_{j+1} = d^{s_j} \geq s_j + \delta_0(d)$, hence $s_j \rightarrow \infty$; by [2, Lemma 2.5] the divergence self-improves to $s_{J_1+k} \geq 2^k s_{J_1}$ and $\sum_j 1/s_j < \infty$. In particular Hypothesis 2.4 of [1] holds unconditionally.

Proof. $x \mapsto d^x - x$ is convex with critical point $x_* = -\ln \ln d / \ln d$, where the minimum equals $d^{x_*} - x_* = (1 + \ln \ln d) / \ln d$. For $d > e^{1/e}$ one has $\ln \ln d > -1$, so the minimum is positive. The remaining statements are immediate (the inf in s_j is attained at $\theta = 0$ since T_d is increasing). $\square$

4 The Stoppregel: unconditional finite-chain positivity

The following lemma re-reads the tail computation of [2, Lemma 6.3] *unconditionally*: the downward induction produces the positivity of the chain instead of assuming it, at the price of stopping J_0 levels above the bottom.

Lemma 2 (Stopped channel). *Let $b, d > e^{1/e}$ and let*

$$J_0 = J_0(b, d) := \min\{j \geq 0 : s_j \geq 2C_0\} \quad (\text{finite by Lemma 1}).$$

Then for every $n \geq J_0 + 1$ and $\theta \in [0, 1]$ the chain values $W_k^{(n)}(\theta)$, $0 \leq k \leq n - J_0$, are defined and satisfy, for $1 \leq k \leq n - J_0$,

$$W_k^{(n)}(\theta) = \alpha T_d(n - k + \theta)(1 + \sigma_k), \quad \sigma_1 = 0, \quad |\sigma_k| \leq \frac{C_0}{s_{n-k}} \leq \frac{1}{2} \quad (k \geq 2). \quad (1)$$

In particular $W_k^{(n)} > 0$ for $0 \leq k \leq n - J_0$: hypothesis (R') holds unconditionally for the chain stopped at level J_0 .

Proof. $W_0 = T_d(n + \theta) > 0$. The first peel is exact: $W_1 = L_b(d^{T_d(n-1+\theta)}) = \alpha T_d(n - 1 + \theta)$, i.e. (1) with $\sigma_1 = 0$. Assume (1) for some k with $j := n - k \geq J_0 + 1$. Then $|\sigma_k| \leq \frac{1}{2}$, so $W_k > 0$ and $L_b(W_k)$ is defined; using $L_b(T_d(j + \theta)) = \alpha T_d(j - 1 + \theta)$ and additivity of L_b on positive factors,

$$W_{k+1} = L_b(\alpha) + \alpha T_d(j - 1 + \theta) + L_b(1 + \sigma_k) = \alpha T_d(j - 1 + \theta)(1 + \sigma_{k+1}),$$

$$\sigma_{k+1} = \frac{L_b(\alpha) + L_b(1 + \sigma_k)}{\alpha T_d(j - 1 + \theta)}, \quad |\sigma_{k+1}| \leq \frac{|L_b(\alpha)| + \ln 2 / \ln b}{\alpha s_{j-1}} = \frac{C_0}{s_{j-1}} \leq \frac{1}{2},$$

the last step because $j-1 \geq J_0$. The induction proceeds while $n-k \geq J_0+1$, i.e. up to $k = n - J_0$. $\square$

Corollary 3 (Stopped evaluation and stopped tails). *Define the stopped tails and the stopped phase by*

$$D_n^{\text{stop}}(\theta) := W_{n-J_0}^{(n)}(\theta) = \alpha T_d(J_0 + \theta)(1 + \sigma_{n-J_0}), \quad \Phi_n^{\text{stop}}(\theta) := (n - J_0) + A_b(D_n^{\text{stop}}(\theta)) - (n + \theta).$$

These are defined for all $n \geq J_0 + 1$ and $\theta \in [0, 1]$ without any channel hypothesis. Wherever the full chain of [1] exists, the global Abel equation gives $A_b(D_n^{\text{stop}}) = A_b(D_n) + J_0$, so $\Phi_n^{\text{stop}} = R_n$; the stopped object extends the definition to all admissible pairs. All tail estimates of [2, §6] hold verbatim for the index range $j \geq J_0$.

Remark 4. *Numerically, Corollary 3 is precisely the evaluation ladder used throughout the measurement campaign (peel down to a level with $T_d \approx 10^4$, hand the remainder to slog_b): the Stoppregel is not a workaround but the definition, and (1) doubles as its certified error control, cf. [2, Rem. 7.5(ii)]. For the standard pairs $J_0 \in \{1, 2\}$; even at the edge pair $(e, 1.46)$ one has $J_0 = 23$ (Table 1).*

5 Offset alignment of the limit channel

For the limit objects the channel condition concerns $E_j(\theta) = T_b(j + \tilde{H}(\theta))$, $j \geq 1$ (cf. [2, (11)–(12)]): $E_j > 0$ is what makes the j -th logarithm real. Since $T_b > 0$ exactly on $(-1, \infty)$ and T_b is increasing, the correct statement is an *offset* one.

Lemma 5 (Offset channel). *Assume the phase-law package (P) of [2] for (b, d) , and let*

$$m_\star := \inf_{\theta} \tilde{H}_{b,d}(\theta) \geq \mu_{b,d} - \|P_{b,d}\|_\infty \geq \mu_{b,d} - \sum_{k \geq 1} A_k(b, d), \quad s := \max(0, \lceil -1 - m_\star \rceil).$$

Then $E_j(\theta) > 0$ for all $j \geq s + 1$ and all θ , with margin

$$E_j(\theta) \geq T_b(j - s + (s + m_\star)) > T_b(-1) = 0, \quad j \geq s + 1.$$

Moreover $D^{(s)} := T_b(s + \tilde{H})$ is defined (argument > -2 requires only $s + m_\star > -2$, weaker still), and the entire structure of [1, 2] — endpoint identities, derivative product, groupoid and antisymmetry statements — holds for the s -shifted objects by shift-equivariance of the tetration equation.

Proof. $j + \tilde{H} \geq j - s + (s + m_\star)$ and $s + m_\star > -1$ by the definition of s ; monotonicity and positivity of T_b on $(-1, \infty)$ give the claim. Shift-equivariance: replacing $\tilde{H}$ by $s + \tilde{H}$ replaces D by $D^{(s)} = T_b^\circ$ -transported one level per unit shift, and every identity used in [1, 2] is a consequence of $T(x + 1) = a^{T(x)}$, which is invariant under integer shifts of the height coordinate. $\square$

Example 6 (The unshifted reading fails for skew pairs). For $(b, d) = (10, 2)$: $\mu = -2.26496\dots$, $\|P\|_\infty \leq 4.7 \cdot 10^{-4}$, so $\inf \tilde{H} \leq -2.264 < -2$: even $D = T_b(\tilde{H})$ is undefined on part of the period, and $E_1 = T_b(1 + \tilde{H})$ takes negative values; with $s = 2$ everything is restored, with margin $s + m_\star + 1 = 0.7346$. For $(e, 1.5)$: $\mu = -9.7763\dots$, $s = 9$, margin 0.2234. The measured phase law for these pairs (validated to 10^{-38} antisymmetry in the present campaign) is throughout the phase law of the offset objects.

Remark 7 (Hub anchoring). $\mu_{b,d}$ — hence s — is computable a priori from the hub column $m(\cdot)$ via $\mu_{b,d} = m(d) - m(b) + \Delta^{(2)} + \Delta^{(3)} + \dots$ (validated to $\sim 10^{-15}$ absolute in this campaign), and $\|P\|_\infty$ from the first few A_k plus the universal decay law. A conversion engine can therefore fix the offset before evaluating anything.

6 Numerical certification

Table 1 shows an excerpt of the certification (all 45 measured pairs pass; script `foundation_cert.py`, data at 30–170 digits). Margins are $s + m_\star + 1$ with the rigorous mode bound for $m_\star$.

pair	$\mu_{b,d}$	$\sum A_k$	s	margin	C_0	J_0
$(e, 2)$	−1.128404	$1.5 \cdot 10^{-4}$	1	0.8715	1.529	2
$(2, e)$	+1.128404	$1.5 \cdot 10^{-4}$	0	2.1283	1.060	1
$(10, 2)$	−2.264961	$4.7 \cdot 10^{-4}$	2	0.7346	2.732	3
$(2, 10)$	+2.264961	$4.6 \cdot 10^{-4}$	0	3.2645	0.822	1
$(3, 5)$	+0.578835	$1.5 \cdot 10^{-4}$	0	1.5787	0.668	1
$(e, 10)$	+1.136557	$3.2 \cdot 10^{-4}$	0	2.1362	0.663	1
$(e, 1.7)$	−2.879554	$2.5 \cdot 10^{-4}$	2	0.1202	2.500	4
$(e, 1.5)$	−9.776277	$3.6 \cdot 10^{-4}$	9	0.2234	3.936	11
$(e, 1.46)$	−21.94334	$3.9 \cdot 10^{-4}$	21	0.0563	4.399	23
$(e, 100)$	+1.600615	$5.7 \cdot 10^{-4}$	0	2.6000	0.482	0

Table 1: Certification excerpt. The margin is governed by the fractional part of μ : near-integer μ produces thin margins (cf. $(e, 1.46)$), a structure that also appears to organize the measured singularity abscissae $x^*(b, d)$ of the strip analysis.

7 What remains, and the differentiability building block

With Lemmas 1–5 the conditional content of [1, 2] reduces to the convergence statements of the phase law itself (Theorem 5.6/5.7 of [1]) formulated for the stopped chain — estimates that live entirely in the index range $j \geq J_0$ where (1) is unconditional — plus one inequality per pair (the mode-certified $m_\star$ -bound of Lemma 5), which the hub decomposition makes checkable a priori.

Independently, the response-kernel program of the present campaign establishes numerically, to 24+ digits and by two independent routes (complex-step differentiation of the Kneser engine through Paulsen holomorphy; ε -stencils of the base family), that the Kneser family is differentiable in the base parameter $\beta = \ln \ln b$, with kernel $V = \partial_\beta A_b$ characterized by the selection principle (cohomological equation, normalization $V(1) = 0$, prescribed fixed-point singularity with $dL/d\beta = \ln b L^2/(1 - \lambda)$, and reality). This is the second building block Remark 7.4 of [2] asks for; the constructive (engine-free) realization of the selection principle is the subject of the Riemann–Hilbert program documented in the research log.

Authorship and AI Contribution Statement

This paper is part of a five-paper series produced in a directed human–AI collaboration. The author conceived the mixed-base phase program, built and maintains the high-precision tetration engine used for every computation in the series (a fork of Levenstein’s `fatou.gp`; engine, data and scripts at <https://github.com/Lightrunnerwastaken/gp-tetration>), and directed the research: posing the questions, setting priorities between rounds, auditing results, and deciding which claims met the acceptance gates. Substantial parts of the mathematical development and of the numerical work were produced by AI systems under that direction: Anthropic’s Claude in a theory and review role, an autonomous research agent operating the numerical pipeline, and OpenAI’s ChatGPT, which contributed to the original versions of Papers I and II and to manuscript review. In line with prevailing journal policies, AI systems are not listed as authors; the author assumes full and sole responsibility for all statements and proofs, and readers should weigh this provenance in their assessment.

References

- [1] J. Justus, *A Phase Law for Mixed-Base Tetration: Logarithmic Tails, Cocycle Structure, and Singular Asymptotics*, Paper I of the mixed-base tetration series, preprint, July 2026. <https://github.com/Lightrunnerwastaken/gp-tetration>
- [2] J. Justus, *The Mixed-Base Tetration Phase Map Is a Circle Diffeomorphism*, Paper II of the mixed-base tetration series, preprint, July 2026. <https://github.com/Lightrunnerwastaken/gp-tetration>
- [3] J. Justus, *The Base-Derivative of the Kneser Family: Selection Principle, Uniqueness, and Existence via a Riemann–Hilbert Problem*, Paper V of the mixed-base tetration series, preprint, July 2026. <https://github.com/Lightrunnerwastaken/gp-tetration>
- [4] H. Kneser, *Reelle analytische Lösungen der Gleichung $\varphi(\varphi(x)) = e^x$* , J. reine angew. Math. 187 (1950).
- [5] W. Paulsen, S. Cowgill, *Solving $F(z+1) = b^{F(z)}$ in the complex plane*, Adv. Comput. Math. 43 (2017).
- [6] W. Paulsen, *Tetration for complex bases*, Adv. Comput. Math. 45 (2019).